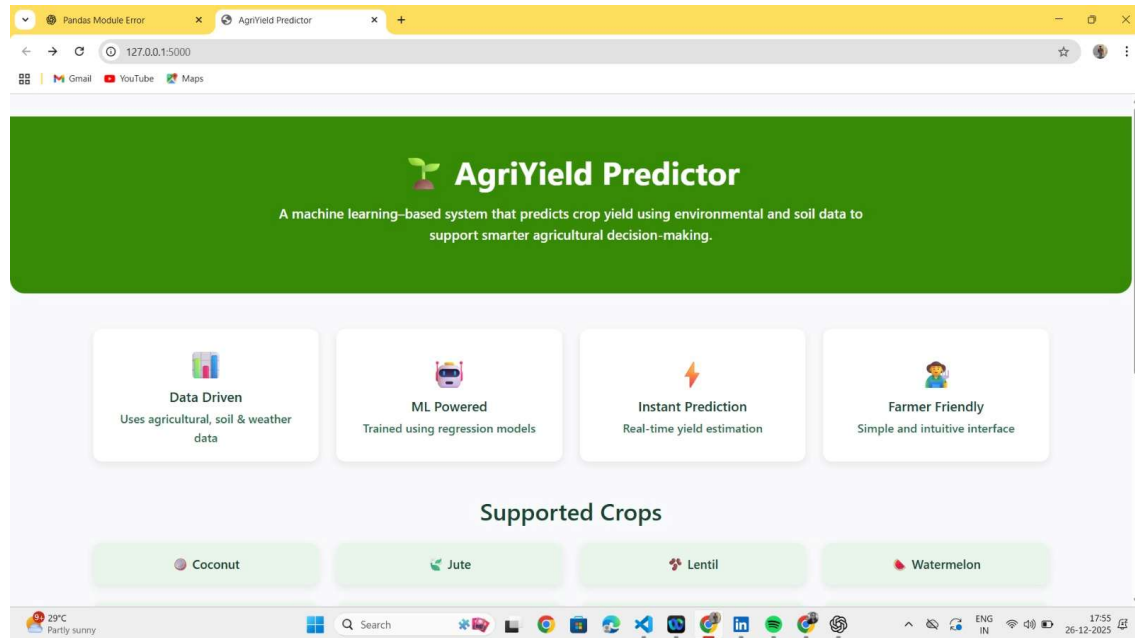


Week 7 – Web UI Development & Real-Time Prediction

Objective

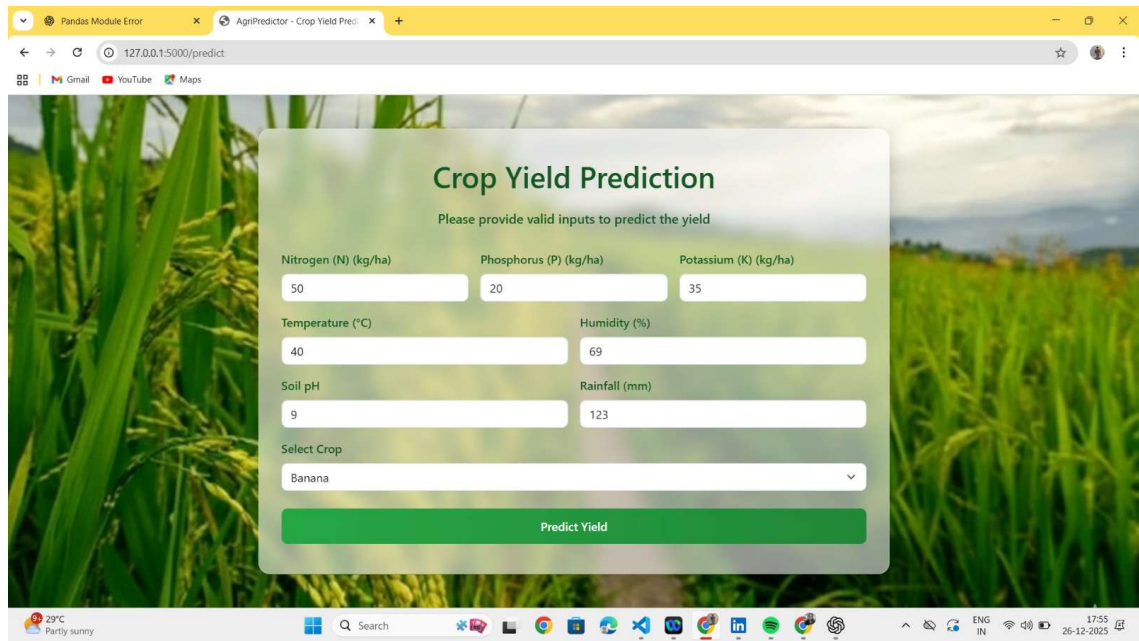
The objective of this week is to develop a simple web-based user interface that allows users to input agricultural parameters and receive crop yield predictions in real time.



Web UI Design

A user-friendly web interface was developed with the following features:

- Tabular or form-based input fields for parameters such as:
 - Crop type
 - Soil type
 - Rainfall
 - Temperature
 - NPK levels
- Input validation to ensure correct data entry



Yield Prediction

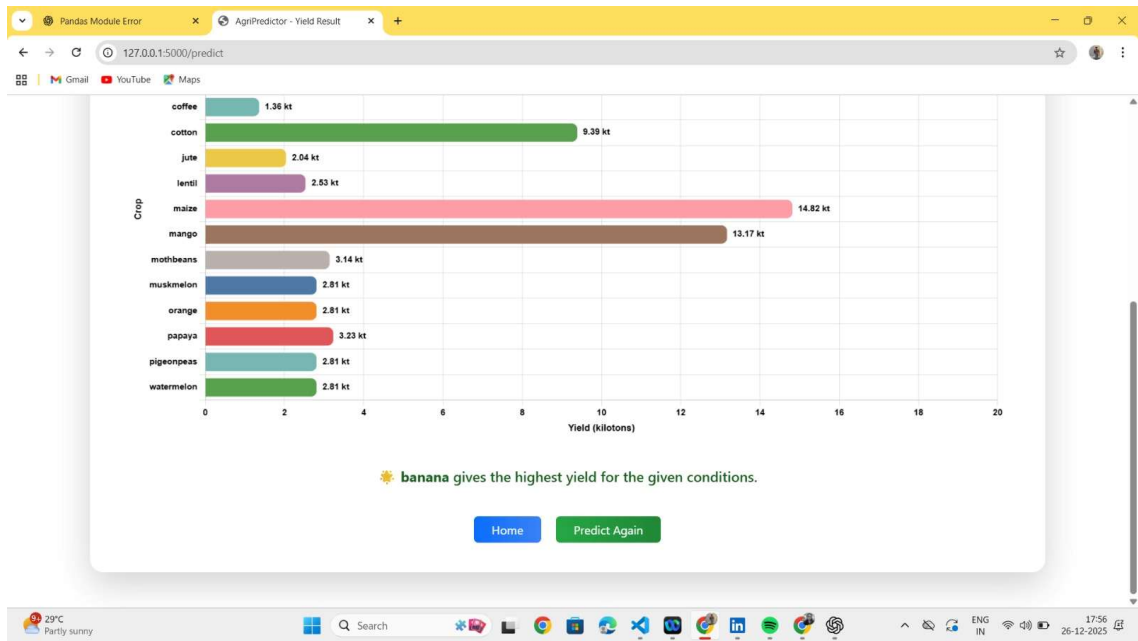
- The trained machine learning model was integrated with the web application.
- Upon submission of input data, the system predicts the crop yield.
- Prediction results are displayed clearly on the UI.

Confidence Interval Estimation

- Confidence intervals were included to represent prediction uncertainty.
- The range helps users understand best-case and worst-case yield estimates.

Real-Time Testing & Validation

- The system was tested using real-time sample inputs.
- Predictions were validated against known values to check consistency.
- Performance and response time were analyzed.



Outcome of Week 7

- Functional web UI created for crop yield prediction
- Real-time model inference successfully implemented
- Confidence intervals added to improve prediction reliability